Day01_Poll_Questions

1.
Consider an array of 10 elements, what will be the time complexity, if the element is found on the 7th position using linear search?

A) O (n)
B) O (1)
C) O (n/2)
D) O (log n)

Answer: A

2.
If the Searching is performed on an array which is sorted in the descending order, which element will give the best-case time complexity?

A) Element at first position
B) Element at last position
C) Element at mid position
D) None of the above

Answer: C

3.
Consider an array 11,22,33,44,55,66. After how many comparisons will we find the element 22 using binary search?

A) 1
B) 2
C) 3
D) 4

Answer: C

4.
What is the Space complexity Of Linear and Binary Search respectively?

A) O(1) O(n)
B) O(n) O(1)
C) O(n) O(n)
D) O(1) O(1)

Answer: D

5.
What is Total Space Complexity?

A) Total Input Space
B) Total Auxiliary Space
C) Input + Auxiliary Space
D) Space Taken by the code in the text section.

Answer: C

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Day02_Poll_Questions

1.
Insertion sort algorithm allows comparison from which index?

A) 0th index
B) n-1 index
C) 1st index
D) mid index

Answer: C

2.
Consider an array: int arr [5] = {25,20,35,12,40}
What will be the sequence of the array after the 3rd pass using bubble sort.

A) 12 20 25 35 40
B) 20 25 12 35 40
C) 20 12 35 25 40
D) 25 20 12 35 40

Answer: A

3.
Which of the following algorithms works well in the worst case scenario?

A) Insertion Sort
B) Quick Sort
C) Bubble Sort
D) Merge Sort

Answer: D

4.
Which of the following algorithms does not follow the divide and conquer technique?

A) Quick Sort
B) Insertion Sort
C) Merge Sort
D) Binary Search

Answer: B

5.
Which of the following sorting algorithms uses extra space on the stack for the stack frames?

A) Selection Sort
B) Bubble Sort
C) Insertion Sort
D) Quick Sort

Answer: D

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1.
Consider a stack of 5 elements. What will be the output if we push an element 50 at the 5th index?

A) 50 is inserted into the stack
B) No output
C) Stack Overflow Runtime Error
D) Stack Underflow Runtime Error

Answer: C

2.
Consider the following operations performed on the stack of 5.
Push (11), Push (66), Push (22), Peek (), Pop (), Push (55), pop (), Peek (), Push (33), Push (44), Push (55).
What will be the element at the 2nd index.

A) Error: Duplicate values :55 cannot be inserted twice
B) 22
C) 66
D) 33

Answer: D

3.
How are the operators handled when we convert an Infix Expression to Postfix Expression?

A) Operators are Appended into the final postfix expression.
B) Compared to one another for precedence and then pushed on stack.
C) Pushed on stack from left to right one by one.
D) Operators are ignored.

Answer: B

4.
Evaluate the below integer postfix expression using stack.

"8 4 - 3 7 5 1 * - / +"

A) 5
B) 5.5
C) 4
D) 6

Answer: A

5.
Convert the following Infix expression to Prefix:
A * B + (C - D / E) * (F + G)

A) + * A B * / - C D E + F G

B) + * A B * - C / D E + F G

C) + * * A B - C / D E + F G

D) * + A B * - C / D E + F G

Answer: B

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Day04_Poll_Questions

1.
What is the time complexity to insert an element into the queue of 'n' elements.

A) O(1)
B) O(n)
C) O(log n)
D) O(n^2)

Answer: A

2.
What is the queue full condition for the circular queue of n elements, if front and rear are set to -1.

A) Front == Rear
B) Front == (Rear + 1) % n
C) Front > Rear
D) Front = Rear = -1

Answer: B

3.
Which of the following is NOT true about circular queues?

A) It Follows First In First Out Mechanism.
B) Insertion Is done from the Rear End and deletion is done from the Front End.
C) The memory space of the queue can be Re-utilised.
D) Insertion is done from the Front End and deletion is done from the Rear End.

Answer: D

4.
Which of the following are not the queue applications?

A) Customer Service Calls
B) Disk scheduling
C) Traffic Management systems
D) Undo / redo Operations

Answer: D

5.
Implementation of Linear Queue can be done using:

A) Only Static Implementation.
B) Only Dynamic Implementation.
C) Both Static and Dynamic Implementation.
D) Not Sure.

Answer: C

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1.
In a singly Linked List with N nodes, how many links are there?

A) n
B) n-1
C) n+1
D) n^2

Answer: B

2.
Consider the linked list has 4 nodes. What will be the value of n returned from the function?

```
struct Node {
   int data;
   struct Node* next;
};

int count (struct Node* head) {
   int n = 0;
   while (head) {
      n++;
      head = head->next;
   }
   return n;
}
```

A) 4
B) 5
C) 6
D) Infinite Loop

Answer: A

3.
What does the below function do?

```
void display (struct Node* head) {
   if (head == NULL)
 return;
   display(head->next);
   printf ("%d ", head->data);
}
```

A) Prints the list in original order.
B) Prints the list in reverse order.
C) Displays the Last node data.
D) Displays the first node data.

Answer: B

4.
Consider the Singly Linear Linked List with the following nodes.
11 -> 22 -> 33 -> 44
What will the following code print?

```
void PrintList (struct Node* head)
{
   while (head->next != NULL)
   {
      printf ("%d ", head->data);
      head = head->next;
   }
}
```

A 11 22 33
B) 11 22 33 44
C) 44 33 22 11
D) Infinite Loop

Answer: A

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5.
What is the exit condition to stop traversal in a singly linked list, if the trav pointer begins from the head?

A) while (trav->next != NULL)
B) while (trav == NULL)
C) while (trav != NULL)
D) while (trav->next == NULL)

Answer: C

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Day06_Poll_Questions

1.
Which of the following are the advantages of doubly linear linked list over singly Linear linked list with the head and tail pointer in both the lists?

I) Requires less memory space.
II) Can traverse backwards.
III) Easier to delete the last node in O(1) time.
IV) Easier to add the node at the last position in O(1) time.

A) Only II
B) Only III
C) Only III and IV
D) Both II and IV

Answer: D

2.
How many pointers are updated if we delete a node in the middle of a doubly linked list?

A) Only the next pointer is updated.
B) Only the prev pointer is updated.
C) Both next and prev pointers are updated.
D) No pointers are updated.

Answer: C

3.
What will be the sequence of nodes if the following functions are called?
add_first_node (10)
add_last_node (20)
add_first_node (30)
add_Last_node (40)
reverse_display ()

A) 10->20->30->40
B) 40->30->20->10
C) 30->10->20->40
D) 40->20->10->30

Answer: D

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4.
Select the correct option to check if the linked list has only one node.

A) if (head == NULL)
B) if (head->next == NULL)
C) if (head->prev == NULL)
D) if (head != NULL)

Answer: B

5.
Select the correct option to print the node data in the doubly linear linked list.

A)
```
struct node *trav = head;
do{
    printf ("%d “, trav->data);
    trav = trav->next;
} while(trav->next != head);
```

B)
```
struct node *trav = head;
do{
    printf ("%d ", trav->data);
    trav = trav->next;
} while(trav != head);
```

C)
```
struct node *trav = head;
do{
    printf ("%d ", trav->data);
    trav = trav->next;
} while(trav->next != NULL);
```

D)
```
struct node *trav = head;
do{
    printf ("%d ", trav->data);
    trav = trav->next;
} while(trav != NULL);
```

Answer: D

1.
What is the time complexity to add a node at the first position in the singly circular Linked List if the list is not empty?

A) O(1)
B) O(n)
C) O(log n)
D) None of the above

Answer: B

2.
Select the correct option to attach a new node in an empty circular linked list.
struct node *new_node = (struct node*) malloc (sizeof (struct node));

I) head = new_node;
new_node->next = NULL;

II) head = new_node;
new_node->next = head;

III) head = new_node;

IV) head = new_node;
new_node->next = new_node;

A) Only I
B) Only II
C) Only III
D) Both I and III
E) Both II and IV

Answer: E

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3.
Select the correct option to print the node data in the Circular linked list.

A)
```
struct node *trav = head;
do{
  printf ("%d ", trav->data);
  trav = trav->next;
} while(trav->next != head);
```

B)
```
struct node *trav = head;
do{
  printf ("%d ", trav->data);
  trav = trav->next;
} while(trav != head);
```

C)
```
struct node *trav = head;
do{
  printf ("%d ", trav->data);
  trav = trav->next;
} while(trav->next != NULL);
```

D)
```
struct node *trav = head;
do{
  printf ("%d ", trav->data);
  trav = trav->next;
} while(trav != NULL);
```

Answer: B

4.
What will be the output of the following code if the circular linked list contains 3 nodes?

```
struct Node* tail = head;
while (tail->next != head) {
   tail = tail->next;
}
printf ("%d", tail->data);
```

A) Prints the first nodes data
B) Prints the last nodes data
C) Prints all the nodes data
D) Infinite Loop

Answer: B

5.
Which of the following Linked List is the most efficient in terms of time?

A) singly Liner Linked List
B) Singly Circular Linked List
C) Doubly Linear Linked List
D) Doubly Circular Linked List

Answer: D

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Day08_Poll_Questions

1.
Which of the following is true about a Binary Search Tree?

I) A Tree where all the nodes are arranged in ascending Order
II) A tree where each node has a degree 2
III) A Tree where each node can have a degree 0,1 or 2
IV) A tree where the left child is less than the root and the right child is greater or equal to the root.

A) Only II
B) Only III
C) Both II and IV
D) Both III and IV

Answer: D

2.
What is the formula for the Balance Factor of a node?

A) Height of Right SubTree - Height of Left SubTree
B) Height of Left SubTree - Height of Right SubTree
C) Max (height of left, height of right tree) + 1
D) -1

Answer: B

3.
What is a cycle in a graph?

A) A sequence of edges with no repetition.
B) A path where the starting and ending nodes are the same.
C) A disconnected component.
D) A loop on a node.

Answer: B

4.
What is a complete graph?

A) Every vertex has no edges.
B) Vertices are divided into 2 disjoint sets.
C) No cycles.
D) Every vertex is connected to every other vertex.

Answer: D

5.
What is the load factor in a hash table?

A) Number of collisions
B) Size of the hash table
C) Ratio of the number of elements to the size of the table
D) Number of keys

Answer: C

6.
What is the worst-case time complexity for searching in a hash table with collisions?
A) O (1)
B) O (log n)
C) O (n^2)
D) O (n)

Answer: D

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